SSC MTS 2024 question paper with answers

16 October 2024 · Shift 1

79 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 11 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 14 questions

Q.1

The individual weights (in kg) of six boys are 55, 56, 57, 60, 59 and 55. A new boy joins the group, and the mean weight of the group becomes 62 kg. Find the weight of the new boy (in kg).

(A) 94
(B) 90
(C) 92
(D) 82

Q.2

The average age of four boys Shikhar, Mukesh, Anil, and Anish is 25 years. If the ages of Shikhar, Mukesh, and Anil are 15, 25, and 29 years, then what is the age (in years) of Anish?

(A) 29
(B) 32
(C) 30
(D) 31

Q.4

A man on a tour of 400 km, travels the first 120 km at 80 km/h and the next 150 km at 75 km/h. If he must spend less than 5 hours in the tour, what is the minimum speed at which he has to travel for the remaining distance?

(A) 88.56 km/h
(B) 84.72 km/h
(C) 86.66 km/h
(D) 90.21 km/h

Q.5

Find the simple interest (in ₹) on ₹3000 as a sum borrowed at 7% per year rate of interest for 6 years.

(A) 1220
(B) 1260
(C) 1200
(D) 1300

Q.6

Ajit walks around a circular field at the speed of two rounds per hour, while Mohit runs around it at the speed of 10 rounds per hour. They start in the same direction from the same point at 6.30 a.m. Find the time at which they cross each other.

(A) 06 : 40 : 00
(B) 06 : 37 : 50
(C) 06 : 37 : 30
(D) 06 : 37 : 20

Q.7

There are 98 women in a group and their average weight is 52 kg. If 7 women of average weight 60 kg leave the group and two new women of average weight 80 kg join the group, then what will be the new average weight of women in the group now?

(A) 54 kg
(B) 50 kg
(C) 56 kg
(D) 52 kg

Q.9

Find the simple interest (in ₹) on ₹3500 as a sum borrowed at 4% per year rate of interest for 6 years.

(A) 840
(B) 880
(C) 800
(D) 780

Q.10

Which of the following is true about average?

(A) The average of six consecutive even integers is an even natural number
(B) The average of six consecutive odd integers is an odd natural number.
(C) The average of six consecutive odd integers is an even integer.
(D) The average of six consecutive odd integers is an odd integer.

Q.11

Simple interest on a given sum for 3 years at a rate of 17% per year is ₹7,140 less than the simple interest on the same sum for 5 years. Find the sum.

(A) ₹21,000
(B) ₹22,000
(C) ₹23,000
(D) ₹20,000

Q.12

The number of schools in a city is directly proportional to the population of the city. In 2020, there were 125 schools and the population of the city was 1,80,000. In 2023, the number of schools in the city is 180. What is the percentage increase in the population of the city from 2020 to 2023?

(A) 44%
(B) 46%
(C) 42%
(D) 45%

Q.13

By selling an article for ₹19, a dealer makes a profit of 25%. By how much should he increase his selling price (in ₹) so as to make a profit of 50%?

(A) 3.8
(B) 5.8
(C) 6.8
(D) 4.8

Q.15

The marked price of a table is ₹10,500, and it was sold for ₹9,450. Find the percentage discount for this transaction.

(A) 9.5%
(B) 10%
(C) 10.5%
(D) 9%

Q.17

A teacher wanted certain groups of students with exactly 10 or12 or 15 members. What minimum number of students must be there in the class to make the required groups?

(A) 60
(B) 120
(C) 37
(D) 150

Q.19

Find the total surface area of hemisphere of radius 6.3 cm.

(A) 3,742 cm^2
(B) 3,524 cm^2
(C) 3,777 cm^2
(D) 374.2 cm^2

Reasoning Ability and Problem Solving 15 questions

Q.1

ADKR is related to CBMP in a certain way based on the English alphabetical order. In the same way, GXQL is related to IVSJ. To which of the following is OPYD related, following the same logic?

(A) QANB
(B) BANQ
(C) QNAB
(D) QBAN

Q.3

What will come in the place of '?' in the following equation if '+' and '×' are interchanged and '–' and '÷' are interchanged? 100 – 4 ÷7 + 5 × 12 = ?

(A) 2
(B) 5
(C) 6
(D) 4

Q.4

BI54 is related to XE64 in a certain way. In the same way, DK76 is related to ZG86. To which of the following is FM43 related, following the same logic?

(A) EB35
(B) IB53
(C) BE35
(D) BI53

Q.6

What should come in place of the question mark (?) in the given series? 247 258 280 313 357 ?

(A) 388
(B) 401
(C) 426
(D) 412

Q.7

In a certain code language, 'HARD' is coded as '1596' and 'CARD' is coded as '5689'. What is the code for 'C' in the given code language?

(A) 9
(B) 5
(C) 6
(D) 8

Q.8

Select the set in which the numbers are related in the same way as are the numbers of the following sets. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into their constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) (34, 22, 63) (61, 76, 144)

(A) (86, 13, 96)
(B) (26, 83, 16)
(C) (45, 41, 93)
(D) (3, 22, 61)

Q.9

What should come in place of the question mark (?) in the following series based on the English alphabetical order? DKO, GJL, JII, MHF ,?

(A) PGC
(B) QGB
(C) PGB
(D) QGC

Q.10

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusion(s) logically follow(s) from the statement(s). Statements: All ants are balls. All ants are dolls. Conclusions: (I) Some balls are dolls. (II) All dolls are ants.

(A) Both conclusions (I) and (II) follow.
(B) Only conclusion (I) follows.
(C) Only conclusion (II) follows.
(D) Neither conclusion (I) nor (II) follows.

Q.11

A, B, C, D, E and F are sitting around a circular table facing the centre (but not necessarily in the same order). F sits third to the left of A. D sits to the immediate left of A. C sits second to the right of A. Only E sits between A and C. When counted from right of A, who sits to the immediate left of B?

(A) E
(B) A
(C) F
(D) C

Q.12

What will come in place of the question mark (?) in the following equation, if '+' and '–' are interchanged and '×' and '÷' are interchanged? 69 – 36 × 6 ÷ 12 + 5 = ?

(A) 130
(B) 129
(C) 128
(D) 136

Q.14

13 is related to 38 following a certain logic. Following the same logic, 15 is related to 44. To which of the following is 26 related, following the same logic? (NOTE : Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding / subtracting / multiplying, etc., to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 86
(B) 77
(C) 68
(D) 71

Q.15

P, Q, R, S, T and U are sitting around a circular table facing the centre (but not necessarily in the same order). R is sitting to the immediate left of P and immediate right of T. Q is sitting to the immediate left of T. U is sitting to the immediate left of Q. P is sitting to the immediate left of S. Who is sitting to the immediate right of S?

(A) T
(B) U
(C) Q
(D) R

Q.16

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: Some pizzas are burgers. Some pizzas are sausages. Conclusions: (I) At least some sausages are burgers. (II) Some sausages are pizzas.

(A) Both conclusions (I) and (II) follow
(B) Only conclusion (II) follows
(C) Neither conclusion (I) nor (II) follows
(D) Only conclusion (I) follows

Q.17

In a certain code language, 'joy salt area' is coded as 'sd vt de' and 'area kit care' is coded as 'vt fu pv'. How is 'area' coded in the given language?

(A) vt
(B) de
(C) fu
(D) sd

Q.18

In a certain code language, 'TASK' is coded as '3762' and 'CAST' is coded as '2436'. What is the code for 'K' in that language?

(A) 6
(B) 4
(C) 2
(D) 7

General Awareness 25 questions

Q.1

Which of the following statements is/are correct according to the Census of India-2011? (a) A person aged seven years or above, who can both read and write with understanding in any language, is treated as literate. (b) A person who can only read but cannot write is not literate.

(A) Only (b) is correct
(B) Only (a) is correct
(C) Neither (a) nor (b) is correct
(D) Both (a) and (b) are correct

Q.2

Which of the following statements are correct about Ribosomes? Statement A: Ribosomes have no membrane. Statement B: Digestive hydrolytic enzymes are present in Ribosomes. Statement C: These are called suicidal bags. Statement D: Ribosomes play an important role in protein synthesis.

(A) Only statements A and C
(B) Only statements A and D
(C) Only statements A and B
(D) Only statements A, B and C

Q.3

Which of the following award did Pandit Ravi Shankar win for four times?

(A) The International Classical Music Awards
(B) The Grammy Awards
(C) The CMA Awards
(D) The BRIT Awards

Q.4

In May 2023, Prime Minister Narendra Modi has been awarded the highest honours by the Pacific island nations of Papua New Guinea and _____.

(A) Kiribati
(B) Nauru
(C) Tuvalu
(D) Fiji

Q.5

The Pradhan Mantri Rojgar Protsahan Yojana (PMRPY) was launched by which Ministry of Government of India in the year 2016?

(A) Ministry of Housing and Urban Affairs
(B) Ministry of Rural Development
(C) Ministry of Finance
(D) Ministry of Labour and Employment

Q.6

Which of the following articles mentions about the Appropriation Bills in the state legislature?

(A) Article 216
(B) Article 209
(C) Article 213
(D) Article 204

Q.7

A molecule with the chemical formula C_2H_6 is a/an _____ molecule.

(A) propane
(B) ethane
(C) methane
(D) butane

Q.8

Who is the recipient of Padma Vibhushan 2000 for reviving the Odissi Classical dance form?

(A) T Balasaraswati
(B) Pandit Jasraj
(C) Hari Prasad Chaurasia
(D) Kelucharan Mohapatra

Q.9

Which of the following is a public sector industry in India?

(A) Bajaj
(B) Hindustan Unilever Limited
(C) Tata Iron and Steel Company (TISCO)
(D) Steel Authority of India Limited (SAIL)

Q.10
Which of the following states launched Mukhya Mantri Vriksh Sampada Yojana in March 2023?

(A) Punjab
(B) Karnataka
(C) Chhattisgarh
(D) Madhya Pradesh

Q.11
Nife layer of earth is situated in the _____.

(A) Mantle
(B) Core
(C) Lithosphere
(D) Crust

Q.12
In which of the following months is the Tuluni festival celebrated in Nagaland by the Sumi tribe?

(A) July
(B) August
(C) June
(D) May

Q.13
Who among the following was the first Law and Justice Minister of independent India?

(A) Jagjivan Ram
(B) BR Ambedkar
(C) Rajkumari Amrit Kaur
(D) John Matthai

Q.14
Which city hosted the XXII Commonwealth Games 2022?

(A) Birmingham
(B) Gold Coast
(C) Glasgow
(D) Delhi

Q.15
Which of the following crops that spread to northern India in the early years of the 17th century was NOT mentioned in the Ain-i-Akbari in the lists of crops in northern India?

(A) Wheat
(B) Arhar
(C) Tobacco
(D) Jowar

Q.16
Which of the following pairs is INCORRECTLY matched?

(A) Function of President - France
(B) Rule of Law - Britain
(C) Concurrent List - Australia
(D) Procedure Established by Law - Japan

Q.17
In tennis, what is a 'love' score?

(A) The highest possible score in a game
(B) The score when the game is tied
(C) The score when a player wins the first point in a game
(D) The score when a player has zero points

Q.18
Which of the following was the first subcontinental empire?

(A) Mauryan Empire
(B) Gupta Empire
(C) Satavahana Empire
(D) Kushan Empire

Q.19
The Indian Bureau of Mines (IBM) was established in the year _____.

(A) 1985
(B) 1948
(C) 1960
(D) 1973

Q.20
What are the three components of the empowerment model in SAPAP of SHGs?

(A) Social mobilisation, economic development, and education
(B) Collective empowerment, capital formation, and poverty eradication
(C) Social mobilisation, capital formation, and capacity building
(D) Public resource allocation, capacity building, and microfinance

Q.21
What cellular structure is responsible for breaking down and recycling cellular components and molecules?

(A) Mitochondrion
(B) Ribosome
(C) Nucleus
(D) Lysosome

Q.22
'Yakshagana', the famous folk dance, is associated with which state?

(A) Manipur
(B) Bihar
(C) Karnataka
(D) Kerala

Q.23
The famous Qila-i-Kuhna (mosque of the old fort) was built by:

(A) Sher Shah
(B) Alauddin Khilji
(C) Akbar
(D) Humayun

Q.24
'Getting A Grip: On My Body, My Mind, My Self' is an autobiography by who among the following tennis icons?

(A) Anna Kournikova
(B) Monica Seles
(C) Steffi Graf
(D) Maria Sharapova

Q.25
The College of Fort William was established at _____ under British rule.

(A) Calcutta
(B) Bombay
(C) Delhi
(D) Madras

English Language and Comprehension 25 questions

Q.1
The following sentence has been split into four segments. Identify the segment that contains a grammatical error. You know that I am fond of reading / and also that all books are my favourite / but you don't know that this is one of / my favourite book.

(A) You know that I am fond of reading
(B) but you don't know that this is one of
(C) and also that all books are my favourite
(D) my favourite book.

Q.2
Select the option that correctly rectifies the underlined spelling error. The worlds you and I belong to are very "defferent".

(A) deffirent
(B) diferent
(C) different
(D) deferent

Q.3
Select the most appropriate synonym of the given word. Vague

(A) Distinct
(B) Dissimilar
(C) Indefinite
(D) Intact

Q.4

Select the most appropriate ANTONYM of the given word. Fluctuate

(A) Mitigate
(B) Steady
(C) Descend
(D) Wobble

Q.5

Select the INCORRECTLY spelt word.

(A) Supperannuation
(B) Accentuation
(C) Suppressant
(D) Phytoplankton

Q.6

Select the grammatically correct sentence from the options given.

(A) Neither the boy nor the mother could identify the man.
(B) Neither a boy nor a mother could identify the man.
(C) Neither the boy nor a mother could identify a man.
(D) Neither a boy nor the mother could identify a man.

Q.7

Select the option that can be used as a one-word substitute for the given group of words. The space or room that is at the top of a building or just below the main roof

(A) Dome
(B) Ceiling
(C) Attic
(D) Stairs

Q.8

Select the most appropriate option that can substitute the underlined segment in the given sentence. We “can observed from” the poll that the award is likely to go to the horror movie director this year.

(A) are observing of
(B) had observe from
(C) have been observing of
(D) can observe from

Q.9

Select the most appropriate idiom to fill in the blank. Getting a sudden promotion and increment in my current profile was like _____.

(A) a bolt from the blue
(B) a fish out of water
(C) a ballpark figure
(D) a foul play

Q.10

Select the most appropriate ANTONYM of the given word. Extensive

(A) Expensive
(B) Abstract
(C) Concrete
(D) Limited

Q.11

Select the option with the correct spelling to replace the underlined word in the given sentence. Development does not take place in a “vaccum”.

(A) vaccim
(B) vacuuem
(C) vakum
(D) vacuum

Q.12

Select the most appropriate option that can substitute the underlined segment in the given sentence. He is the “healthy” person I have ever met in life.

(A) most healthy
(B) more healthy
(C) healthiest
(D) healthier

Q.13

Select the most appropriate ANTONYM of the given word. Amplify

(A) Defy
(B) Contract
(C) Exert
(D) Enlarge

Q.14

The following sentence has been split into four segments. Identify the segment that contains an error. Stephan, however, contrived / to put away / his departure / for eighteen months.

(A) for eighteen months
(B) Stephan,however,contrived
(C) to put away
(D) his departure

Q.15

The following sentence has been divided into parts. One of them may contain an error. Select the part that contains the error from the given options. If you don't find any error, mark 'No error' as your answer. Her decision, / given the complex circumstances, / was too impulsively made.

(A) No error
(B) given the complex circumstances,
(C) was too impulsively made.
(D) Her decision,

Q.16

The following sentence has been divided into parts. One of them may contain an error. Select the part that contains the error from the given options. If you don't find any error, mark 'No error' as your answer. Chetan's room is clean / than Ram's room because / Chetan cleans it more often than Ram.

(A) Chetan's room is clean
(B) than Ram's room because
(C) No error
(D) Chetan cleans it more often than Ram

Q.17

Select the most appropriate option to fill in the blanks. Our _____ has so many magnificent tourist destinations, with unimaginable potential, yet to ____ fully explored.

(A) state; been
(B) country; be
(C) land; has
(D) national; have

Q.18

Select the most appropriate ANTONYM of the given word. Expand

(A) Inflate
(B) Protect
(C) Shrink
(D) Network

Q.19

Select the most appropriate option to fill in the blank. I could make out from my son's voice that he was _____ as his exam did not go well.

(A) rejected
(B) abrogated
(C) confused
(D) sad

Q.20

Select the most appropriate meaning of the given idiom. Once in a blue moon

(A) A pleasant surprise
(B) A frequent event
(C) A rare occurrence
(D) A regular activity

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. She went on a quest for ancient things ___(1)____ could have a soul of their own, something ___(2)___ Aladdin's lamp that was home to a genie. She searched flea markets, met with art dealers and even went to excavation sites. She ___(3)__ searching for something that would have seen some ancient history when an ___(4)___, ___(5)___, ___(6)____ apothecary table caught her attention. It was perhaps owned by a __(7)____, ___(8)___, __(9)____ woman who was a wise medic but was deemed a witch.

Select the most appropriate option to fill in blank number 1.

(A) which
(B) who
(C) whose
(D) that

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. She went on a quest for ancient things ___(1)____ could have a soul of their own, something ___(2)___ Aladdin's lamp that was home to a genie. She searched flea markets, met with art dealers and even went to excavation sites. She ___(3)__ searching for something that would have seen some ancient history when an ___(4)___, ___(5)___, ___(6)____ apothecary table caught her attention. It was perhaps owned by a __(7)____, ___(8)___, __(9)____ woman who was a wise medic but was deemed a witch.

Select the most appropriate option to fill in blank number 2.

(A) like
(B) by
(C) of
(D) upstairs

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. She went on a quest for ancient things ___(1)____ could have a soul of their own, something ___(2)___ Aladdin's lamp that was home to a genie. She searched flea markets, met with art dealers and even went to excavation sites. She ___(3)__ searching for something that would have seen some ancient history when an ___(4)___, ___(5)___, ___(6)____ apothecary table caught her attention. It was perhaps owned by a __(7)____, ___(8)___, __(9)____ woman who was a wise medic but was deemed a witch.

Select the most appropriate option to fill in blank number 3.

(A) is
(B) was
(C) has been
(D) were

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. She went on a quest for ancient things ___(1)____ could have a soul of their own, something ___(2)___ Aladdin's lamp that was home to a genie. She searched flea markets, met with art dealers and even went to excavation sites. She ___(3)__ searching for something that would have seen some ancient history when an ___(4)___, ___(5)___, ___(6)____ apothecary table caught her attention. It was perhaps owned by a __(7)____, ___(8)___, __(9)____ woman who was a wise medic but was deemed a witch.

Select the most appropriate option to fill in blanks numbered 4, 5 and 6 respectively.

(A) antique; heavy; chestnut brown
(B) antique; chestnut brown; heavy
(C) heavy; antique; chestnut brown
(D) chestnut brown; antique; heavy

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. She went on a quest for ancient things ___(1)____ could have a soul of their own, something ___(2)___ Aladdin's lamp that was home to a genie. She searched flea markets, met with art dealers and even went to excavation sites. She ___(3)__ searching for something that would have seen some ancient history when an ___(4)___, ___(5)___, ___(6)____ apothecary table caught her attention. It was perhaps owned by a __(7)____, ___(8)___, __(9)____ woman who was a wise medic but was deemed a witch.

Select the most appropriate option to fill in blanks numbered 7, 8 and 9 respectively.

(A) hunch-backed; old; small

(B) old; small; hunch-backed

(C) small; old; hunch-backed

(D) small; hunch-backed; old

Answer key

Numerical and Mathematical Ability

Q.1 **(C)**	Q.2 **(D)**	Q.4 **(C)**	Q.5 **(B)**	Q.6 **(C)**	Q.7 **(D)**
Q.9 **(A)**	Q.10 **(C)**	Q.11 **(A)**	Q.12 **(A)**	Q.13 **(A)**	Q.15 **(B)**
Q.17 **(A)**	Q.19 **(D)**				

Reasoning Ability and Problem Solving

Q.1 **(C)**	Q.3 **(A)**	Q.4 **(D)**	Q.6 **(D)**	Q.7 **(D)**	Q.8 **(C)**
Q.9 **(A)**	Q.10 **(B)**	Q.11 **(C)**	Q.12 **(D)**	Q.14 **(B)**	Q.15 **(B)**
Q.16 **(B)**	Q.17 **(A)**	Q.18 **(D)**			

General Awareness

Q.1 **(D)**	Q.2 **(B)**	Q.3 **(B)**	Q.4 **(D)**	Q.5 **(D)**	Q.6 **(D)**
Q.7 **(B)**	Q.8 **(D)**	Q.9 **(D)**	Q.10 **(C)**	Q.11 **(B)**	Q.12 **(A)**
Q.13 **(B)**	Q.14 **(A)**	Q.15 **(C)**	Q.16 **(A)**	Q.17 **(D)**	Q.18 **(A)**
Q.19 **(B)**	Q.20 **(C)**	Q.21 **(D)**	Q.22 **(C)**	Q.23 **(A)**	Q.24 **(B)**
Q.25 **(A)**					

English Language and Comprehension

Q.1 **(D)**	Q.2 **(C)**	Q.3 **(C)**	Q.4 **(B)**	Q.5 **(A)**	Q.6 **(A)**
Q.7 **(C)**	Q.8 **(D)**	Q.9 **(A)**	Q.10 **(D)**	Q.11 **(D)**	Q.12 **(C)**
Q.13 **(B)**	Q.14 **(C)**	Q.15 **(C)**	Q.16 **(A)**	Q.17 **(B)**	Q.18 **(C)**
Q.19 **(D)**	Q.20 **(C)**	Q.21 **(D)**	Q.22 **(A)**	Q.23 **(B)**	Q.24 **(A)**
Q.25 **(D)**					

Explanations

Numerical and Mathematical Ability

Q.1 · Answer (C) 92

The six boys weigh 55 + 56 + 57 + 60 + 59 + 55 = 342 kg. Seven boys averaging 62 kg weigh 7 × 62 = 434 kg, so the new boy weighs 434 − 342 = 92 kg.

Q.2 · Answer (D) 31

The four total 4 × 25 = 100 years and the other three 15 + 25 + 29 = 69, so Anish is 100 − 69 = 31.

Q.4 · Answer (C) 86.66 km/h

The first 120 km take 120/80 = 1.5 hours and the next 150 km take 150/75 = 2 hours. The remaining 130 km must take under 1.5 hours, so the speed must be above 130 ÷ 1.5 = 86.666… km/h, which SSC's key gives as 86.66 km/h.

Q.5 · Answer (B) 1260

SI = 3,000 × 7 × 6/100 = ₹1,260.

Q.6 · Answer (C) 06 : 37 : 30

Mohit gains 10 − 2 = 8 rounds an hour on Ajit, so he gains a full round and passes him after 1/8 hour = 7 minutes 30 seconds: at 6:37:30 a.m.

Q.7 · Answer (D) 52 kg

Total weight = 98 × 52 − 7 × 60 + 2 × 80 = 5,096 − 420 + 160 = 4,836 kg for 98 − 7 + 2 = 93 women: 4,836/93 = 52 kg.

Q.9 · Answer (A) 840

SI = 3,500 × 4 × 6/100 = ₹840.

Q.10 · Answer (C) The average of six consecutive odd integers is an even integer.

Six consecutive odd integers are a, a + 2, …, a + 10 with a odd; their average is a + 5, an even integer. (Six consecutive even integers average a + 5 with a even, an odd number.)

Q.11 · Answer (A) ₹21,000

The extra 2 years' interest is ₹7,140: P × 17 × 2/100 = 7,140, so P = 7,140/0.34 = ₹21,000.

Q.12 · Answer (A) 44%

The population rises in the same proportion as the number of schools: from 125 to 180 is a rise of 55/125 = 44%.

Q.13 · Answer (A) 3.8

Cost price = 19/1.25 = ₹15.20. A 50% profit needs a price of 15.20 × 1.5 = ₹22.80, a rise of ₹3.80.

Q.15 · Answer (B) 10%

Discount = 10,500 − 9,450 = ₹1,050, and 1,050/10,500 = 10%.

Q.17 · Answer (A) 60

The class must divide exactly into groups of 10, of 12 or of 15, so the smallest class is LCM(10, 12, 15) = 60.

Q.19 · Answer (D) 374.2 cm²

Total surface area of a hemisphere = $3\pi r^2$ = 3 × 22/7 × 6.3 × 6.3 = 374.22 cm², about 374.2 cm².

Reasoning Ability and Problem Solving

Q.1 · Answer (C) QNAB

The letters move +2, −2, +2, −2: A → C, D → B, K → M, R → P and G → I, X → V, Q → S, L → J. So O → Q, P → N, Y → A (wrapping round) and D → B: QNAB.

Q.3 · Answer (A) 2

After interchanging the signs: 100 ÷ 4 − 7 × 5 + 12 = 25 − 35 + 12 = 2.

Q.4 · Answer (D) BI53
Each letter moves 4 places back and the number rises by 10: B → X, I → E, 54 → 64. So F → B, M → I and 43 → 53: BI53.

Q.6 · Answer (D) 412
The differences are 11, 22, 33, 44, so the next is 55: 357 + 55 = 412.

Q.7 · Answer (D) 8
A, R and D are in both words, as are 5, 9 and 6 in both codes. The letter left in CARD is C and the digit left in its code is 8, so C = 8.

Q.8 · Answer (C) (45, 41, 93)
The third number is the sum of the first two plus 7: 34 + 22 + 7 = 63 and 61 + 76 + 7 = 144. Likewise 45 + 41 + 7 = 93.

Q.9 · Answer (A) PGC
First letters move +3 (D, G, J, M, P), second letters –1 (K, J, I, H, G) and third letters –3 (O, L, I, F, C): PGC.

Q.10 · Answer (B) Only conclusion (I) follows.
Every ant is both a ball and a doll, so some balls are dolls (I). Nothing says every doll is an ant, so II does not follow.

Q.11 · Answer (C) F
Facing the centre, left runs clockwise. F is opposite A, D on A's immediate left, E on its immediate right and C next to E, so clockwise from A the order is A, D, B, F, C, E. F sits to B's immediate left.

Q.12 · Answer (D) 136
After interchanging the signs: 69 + 36 ÷ 6 × 12 – 5 = 69 + 72 – 5 = 136.

Q.14 · Answer (B) 77
The second number is three times the first minus 1: 13 × 3 – 1 = 38 and 15 × 3 – 1 = 44. Likewise 26 × 3 – 1 = 77.

Q.15 · Answer (B) U
Facing the centre, left runs clockwise. R is on P's left and T's right, Q on T's left, U on Q's left and P on S's left, so clockwise the order is P, R, T, Q, U, S. U sits to S's immediate right.

Q.16 · Answer (B) Only conclusion (II) follows
Some pizzas are sausages, so some sausages are pizzas (II). The pizzas that are burgers and those that are sausages need not overlap, so I does not follow.

Q.17 · Answer (A) vt
'area' is the only word in both sentences and 'vt' the only code in both, so area = vt.

Q.18 · Answer (D) 7
T, A and S are in both words, as are 3, 6 and 2 in both codes. The letter left in TASK is K and the digit left in its code is 7, so K = 7.

General Awareness

Q.1 · Answer (D) Both (a) and (b) are correct
The 2011 Census counts a person aged seven or above who can both read and write with understanding in any language as literate; one who can only read is not.

Q.2 · Answer (B) Only statements A and D
Ribosomes have no membrane and are the sites of protein synthesis. Digestive hydrolytic enzymes, and the name 'suicidal bags', belong to lysosomes.

Q.3 · Answer (B) The Grammy Awards
The sitar maestro Ravi Shankar won four Grammy Awards: for 'West Meets East' with the violinist Yehudi Menuhin, 'The Concert for Bangladesh', 'Full Circle: Carnegie Hall 2000' and, after his death, 'The Living Room Sessions Part 1'.

Q.4 · Answer (D) Fiji
In May 2023 Papua New Guinea made Narendra Modi a Grand Companion of the Order of Logohu and Fiji a Companion of the Order of Fiji, each country's highest honour.

Q.5 · Answer (D) Ministry of Labour and Employment
PMRPY, a Ministry of Labour and Employment scheme run through the EPFO, encouraged employers to take on new workers by having the government pay the employer's provident-fund contribution for them.

Q.6 · Answer (D) Article 204
Article 204 provides for the Appropriation Bill in a State Legislature; Article 114 is its counterpart for Parliament.

Q.7 · Answer (B) ethane
C_2H_6 is ethane, the two-carbon alkane; methane is CH_4, propane C_3H_8 and butane C_4H_{10}.

Q.8 · Answer (D) Kelucharan Mohapatra
Guru Kelucharan Mohapatra, who led the revival of Odissi, received the Padma Vibhushan in 2000.

Q.9 · Answer (D) Steel Authority of India Limited (SAIL)
SAIL is a government-owned public sector company; Bajaj, Hindustan Unilever and Tata Steel (TISCO) are private.

Q.10 · Answer (C) Chhattisgarh
Chhattisgarh launched the Mukhya Mantri Vriksh Sampada Yojana on 21 March 2023, World Forestry Day, to encourage farmers to plant commercially valuable trees on their land.

Q.11 · Answer (B) Core
In Suess's division of the Earth's layers, SIAL is the upper crust, SIMA lies below it, and NIFE (nickel and iron) is the core.

Q.12 · Answer (A) July
Tuluni, the main festival of the Sumi Nagas, is celebrated in July.

Q.13 · Answer (B) BR Ambedkar
B. R. Ambedkar was independent India's first Law Minister, in Nehru's first cabinet.

Q.14 · Answer (A) Birmingham
Birmingham, in England, hosted the 22nd Commonwealth Games in 2022.

Q.15 · Answer (C) Tobacco
NCERT notes that tobacco reached the Deccan first and spread to northern India in the early 17th century, and that the Ain-i-Akbari does not list it among the crops of northern India.

Q.16 · Answer (A) Function of President - France
France gave the Constitution the republic and the ideals of liberty, equality and fraternity, not the President's functions. The other pairs are right: the rule of law from Britain, the Concurrent List from Australia and 'procedure established by law' from Japan.

Q.17 · Answer (D) The score when a player has zero points
In tennis 'love' means zero: a player who has not yet won a point in the game is on love.

Q.18 · Answer (A) Mauryan Empire
The Mauryan Empire, founded by Chandragupta Maurya, was the first to bring most of the subcontinent under one rule.

Q.19 · Answer (B) 1948
The Indian Bureau of Mines was set up on 1 March 1948; it now works under the Ministry of Mines.

Q.20 · Answer (C) Social mobilisation, capital formation, and capacity building
SSC's key gives social mobilisation, capital formation and capacity building as the three components of the empowerment model of SAPAP (the South Asia Poverty Alleviation Programme), which organised the poor into self-help groups.

Q.21 · Answer (D) Lysosome
Lysosomes hold hydrolytic enzymes that break down worn-out organelles and other molecules for reuse.

Q.22 · Answer (C) Karnataka
Yakshagana, a folk dance-drama of coastal Karnataka, combines dance, music and dialogue.

Q.23 · Answer (A) Sher Shah
Sher Shah Suri built the Qila-i-Kuhna mosque inside the Purana Qila in Delhi.

Q.24 · Answer (B) Monica Seles
'Getting a Grip' (2009) is the memoir of Monica Seles, the former world number one.

Q.25 · Answer (A) Calcutta
Lord Wellesley founded the College of Fort William at Calcutta in 1800 to train British officials in Indian languages.

English Language and Comprehension

Q.1 · Answer (D) my favourite book.
'One of' takes a plural noun: 'one of my favourite books'.

Q.2 · Answer (C) different
The correct spelling is 'different'.

Q.3 · Answer (C) Indefinite
'Vague' means not clearly stated or defined: 'indefinite'.

Q.4 · Answer (B) Steady
To 'fluctuate' is to rise and fall irregularly; its opposite is to stay 'steady'.

Q.5 · Answer (A) Supperannuation
The correct spelling is 'superannuation', with one 'p'.

Q.6 · Answer (A) Neither the boy nor the mother could identify the man.
The boy, the mother and the man are particular people, so each takes the definite article: 'Neither the boy nor the mother could identify the man.'

Q.7 · Answer (C) Attic
An attic is the space or room just below the roof of a building.

Q.8 · Answer (D) can observe from
A modal such as 'can' is followed by the base form of the verb: 'We can observe from the poll'.

Q.9 · Answer (A) a bolt from the blue
'A bolt from the blue' is a sudden, unexpected event, like an unlooked-for promotion.

Q.10 · Answer (D) Limited
'Extensive' means large in extent or range; its opposite is 'limited'.

Q.11 · Answer (D) vacuum
The correct spelling is 'vacuum': one 'c' and two 'u's.

Q.12 · Answer (C) healthiest
'The … I have ever met' calls for the superlative, and 'healthy' forms it with '-iest': 'the healthiest person'.

Q.13 · Answer (B) Contract
To 'amplify' is to enlarge or increase; its opposite is to 'contract'.

Q.14 · Answer (C) to put away
To postpone is to 'put off': 'contrived to put off his departure for eighteen months'. 'Put away' means to store something or set it aside.

Q.15 · Answer (C) was too impulsively made.
SSC's key marks 'was too impulsively made', presumably for the more natural 'was made too impulsively'. An adverb before the participle ('a hastily made decision') is also acceptable English, so this is a point of style rather than a clear error.

Q.16 · Answer (A) Chetan's room is clean
'Than' needs the comparative: 'Chetan's room is cleaner than Ram's room'.

Q.17 · Answer (B) country; be
A noun is needed after 'Our', and 'yet to' takes the base form: 'Our country has so many … destinations … yet to be fully explored.'

Q.18 · Answer (C) Shrink
The opposite of 'expand' is 'shrink'.

Q.19 · Answer (D) sad
His exam had gone badly, so he sounded 'sad'; 'rejected' and 'abrogated' do not fit, and nothing points to confusion.

Q.20 · Answer (C) A rare occurrence
'Once in a blue moon' means very rarely.

Q.21 · Answer (D) that
SSC's key gives 'that', the usual relative pronoun for a clause defining things: 'ancient things that could have a soul of their own'. ('Which' is also used in such clauses, especially in British English; 'who' is for people and 'whose' is possessive.)

Q.22 · Answer (A) like
'Something like Aladdin's lamp': 'like' means similar to.

Q.23 · Answer (B) was
The past continuous fits an action under way when something else happened: 'She was searching … when an … apothecary table caught her attention.'

Q.24 · Answer (A) antique; heavy; chestnut brown
'An' needs a word beginning with a vowel sound, so 'antique' comes first, and colour comes last, next to the noun: 'an antique, heavy, chestnut brown apothecary table'.

Q.25 · Answer (D) small; hunch-backed; old
SSC's key follows the order that puts size, then shape or build, before age, as Cambridge's grammar does: 'a small, hunch-backed, old woman'. (Lists that put age before shape would give 'small, old, hunch-backed'.)

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